

Thunderstorm Pattern Analysis Using Vision Transformers and Optimal Transport for Aviation Operations

Binshuai Wang* and Peng Wei†
George Washington University, Washington, DC, 20052

Thunderstorms pose significant challenges to aviation safety and operational efficiency, requiring advanced tools for accurate analysis and prediction. This research explores a novel framework integrating Vision Transformers (ViTs) and Optimal Transport (OT) theory to analyze large-scale, unlabelled thunderstorm datasets. Trained in an autoregressive manner, the ViT model extracts both spatial structures and temporal evolution across thunderstorm radar images, producing latent feature embeddings that represent thunderstorm dynamics. These embeddings are treated as empirical distributions (or data support in latent space), enabling the application of OT theory to quantify the similarity between thunderstorm sequences. The resulting similarity measures support comparative analysis and similar sequence retrieval. Experimental results demonstrate that this framework effectively identifies structurally and temporally similar thunderstorm events, with promising implications for enhanced pattern recognition, operational decision-making, and risk mitigation in aviation meteorology.

I. Nomenclature

- $\mathbf{I}_t \in \mathbb{R}^{H \times W \times C}$: Input radar image at time step t , with spatial dimensions $H \times W$ and C channels.
- $\mathbf{O}_t \in \mathbb{R}^{H \times W \times C}$: Predicted radar image at time t , produced by the model.
- P : Patch size used to divide the input image into spatial patches.
- H', W' : Height and width of the downsampled feature map obtained after CNN encoding.
- D : Dimensionality of the latent tokens embedding.
- $N_p = H' \cdot W'$: Number of spatial tokens extracted per frame.
- T : Number of frames in the input sequence.
- $N = T \cdot N_p$: Total number of tokens across all frames in the input sequence.
- $\mathbf{Y}_t \in \mathbb{R}^{N_p \times D}$: Output of the CNN encoder for frame t , representing the spatial tokens.
- $\mathbf{Z}_t \in \mathbb{R}^{N_p \times D}$: Output of the transformer for frame t , representing the spatial-temporal tokens.
- α_{ij} : Attention score indicating the importance of token j to token i in the attention mechanism.
- $\mathcal{L}_{\text{recon}}$: Reconstruction loss computed as the mean squared error between predicted and target frames.
- μ_t, ν_t : Empirical token distributions at time t for two sequences being compared.
- $c(x, y) = \|x - y\|_p^p$: Cost function defining the transport penalty between feature vectors x and y .
- $d_W(\mathbf{A}, \mathbf{B})$: Mean Wasserstein distance across all frames between two latent sequences.

II. Introduction

Thunderstorms pose significant challenges to aviation safety and operational efficiency, requiring advanced tools for accurate analysis and prediction. While data on thunderstorms is abundant, it often lacks the adaptability and resolution needed for in-time operations and decision-making in aviation [1–3]. Traditional approaches are limited by their dependence on handcrafted features and lack of scalability for real-time scenarios [4, 5]. Machine learning techniques offer a promising alternative by leveraging data-driven methods to analyze complex weather phenomena [6], yet challenges remain in adapting these models for highly dynamic storm systems with the temporal dimension. This research addresses these limitations by proposing a novel framework that integrates Vision Transformers (ViTs) and Optimal Transport (OT) theory to analyze large-scale, unlabelled thunderstorm datasets. By combining adaptive methods with explainability, this framework aims to uncover patterns and relationships within thunderstorm data, enhancing both meteorological understanding and aviation decision-making.

The remarkable success of transformer-based architectures in machine learning has revolutionized various domains, starting with their introduction in natural language processing (NLP) [7]. Transformers like BERT [8] and GPT

*Ph.D. Student, Department of Computer Science, George Washington University, AIAA Student Member.

†Associate Professor, Department of Mechanical and Aerospace Engineering, George Washington University, AIAA Associate Fellow.

models [9] have achieved groundbreaking results in language understanding and generation tasks. BERT, an encoder-based model, captures bidirectional context by processing input sequences in both directions, excelling in tasks such as question answering and text classification. In contrast, GPT is a decoder-based model designed for autoregressive tasks, predicting the next token in a sequence based on prior context, making it particularly effective for text generation and few-shot learning. These advancements can be traced back to earlier architectures, such as Memory Neural Networks [10], and sequence-to-sequence models [11]. In computer vision, transformers have also been adapted successfully, starting with Vision Transformers (ViTs) [12], which process images by splitting them into patches and applying attention mechanisms. Subsequent advancements, such as Swin Transformers [13] and hybrid models like ConViT [14], have demonstrated the versatility of transformers in capturing spatial-temporal relationships and their applicability to complex datasets, such as thunderstorms.

While Optimal Transport (OT) provides a mathematically principled method for measuring similarities between distributions [15], directly applying it to raw representations or datasets can be ineffective. This is because raw data representations may not align with the latent space where meaningful patterns are better encoded [16, 17]. Thunderstorm patterns represented in high-dimensional feature space might lack coherence without proper alignment, leading to inaccurate similarity measures. To address this, introducing a neural network-based encoder enables mapping raw thunderstorm data into a latent space where similar weather patterns are effectively aligned [18, 19]. By learning embeddings that emphasize shared structural and intensity features, the OT framework can operate on these latent encodings, ensuring that the Wasserstein distance reflects the underlying similarities between storms. This approach enhances comparative analysis and similar thunderstorm retrieval.

The remainder of this paper is organized as follows: Section III reviews background, including elementary knowledge in ViTs, OT theory, and their applications to spatial-temporal data analysis. Section IV outlines the proposed methodology, detailing how the ViT is employed to extract high-dimensional embeddings from thunderstorm data and how OT is used to measure similarities with the Wasserstein distance. Section V presents experimental results, demonstrating the effectiveness of the approach in comparing and analyzing thunderstorm patterns. Finally, Sections VI discusses outcomes, applications to aviation safety, and potential directions for future research.

III. Background

This section presents related work on Vision Transformers (ViTs) and Optimal Transport (OT) theory, emphasizing their applications in spatial-temporal data analysis and probability distribution comparison. We discuss their respective roles in spatial-temporal data modeling and probability distribution alignment, followed by a synthesis of these methods for thunderstorm pattern analysis. Lastly, we provide a review of related literature.

A. Vision Transformers

Vision Transformers (ViTs) extend the transformer architecture [7] to visual domains by treating image patches as sequential tokens. Unlike convolutional neural networks (CNNs), which operate on local receptive fields, Vision Transformers model global spatial relationships through self-attention. This enables them to capture structured, long-range dependencies critical for understanding meteorological patterns [12, 20]. The following subsections outline key components of the Vision Transformer framework, including its tokenization process, transformer architecture, and training paradigms.

1. Patch Embedding and Tokenization

In Vision Transformers, the first step involves dividing each input image into a grid of patches, typically of equal size (e.g., $P \times P$ pixels). Each patch is then flattened and transformed into a fixed-length vector, referred to as a token, through a learnable linear projection. This tokenization process converts the 2D image into a 1D sequence of tokens, which can be processed similarly to words in a sentence by the transformer architecture. To retain spatial structure, positional embeddings are added to each token, allowing the model to distinguish between different patch locations.

While the original ViT design uses uniformly spaced, non-overlapping patches, subsequent variants such as Swin Transformers [13] introduce overlapping and hierarchical patching strategies to enhance the model’s ability to capture fine-grained and multi-scale spatial features.

2. Self-Attention

The self-attention mechanism computes relationships between tokens by mapping each token to three vectors: a query $\mathbf{q}_i$, a key $\mathbf{k}_j$, and a value $\mathbf{v}_j$. The attention score between tokens i and j is computed as:

$$\alpha_{ij} = \frac{\exp(\mathbf{q}_i \cdot \mathbf{k}_j / \sqrt{d_k})}{\sum_{j=1}^N \exp(\mathbf{q}_i \cdot \mathbf{k}_j / \sqrt{d_k})}, \quad (1)$$

where d_k is the dimension of the key vector. The output for token i is a weighted sum of value vectors:

$$\mathbf{y}_i = \sum_{j=1}^N \alpha_{ij} \cdot \mathbf{v}_j. \quad (2)$$

3. Training Paradigms

Vision Transformers can be trained in two paradigms, each aligned with different model architectures and downstream tasks:

- 1) **Autoregressive Training:** Common in decoder-based architectures, this approach trains the model to predict the next token given past tokens [9]. It is particularly suited for temporal forecasting, where the model extrapolates future weather states based on historical sequences.
- 2) **Non-Autoregressive Training:** Encoder-based transformers use objectives such as masked autoencoding [21], where a subset of tokens is hidden and reconstructed, or contrastive learning [22], which encourages discriminative latent representations. These methods are ideal for unlabeled feature extraction and storm clustering.

B. Optimal Transport Theory and Wasserstein Distance

Optimal Transport (OT) theory provides a mathematical framework for comparing probability distributions by minimizing the cost of transforming one distribution $\mu(x)$ into another $\nu(y)$:

$$\text{OT}(\mu, \nu) = \min_{\gamma \in \Gamma(\mu, \nu)} \int_{\mathbb{R}^n \times \mathbb{R}^n} c(x, y) d\gamma(x, y), \quad (3)$$

where $\Gamma(\mu, \nu)$ is the set of joint distributions (couplings) of μ and ν :

$$\Gamma(\mu, \nu) = \{\gamma \geq 0 \mid \int_{\mathbb{R}^n} \gamma(x, y) dy = \mu(x), \int_{\mathbb{R}^n} \gamma(x, y) dx = \nu(y)\}. \quad (4)$$

$\Gamma(\mu, \nu)$ represents the set of couplings $\gamma(x, y)$ satisfying marginal constraints, and $c(x, y)$ represents the transportation cost, often defined as $\|x - y\|_p^p$. OT is particularly effective for handling differences in shape, size, or density between distributions, making it widely applicable in clustering [23] and meteorological pattern analysis.

Direct computation of OT involves solving a linear programming (LP) problem, where distributions are discretized into finite samples, and transportation costs are minimized. However, this approach becomes computationally prohibitive for large datasets. To improve scalability, Sinkhorn's algorithm regularizes the optimization problem with an entropy term, significantly reducing computational complexity [24]. These computational advances have made OT accessible for real-time and large-scale applications, such as generative modeling and geospatial analysis.

Several extensions of OT have been developed to handle specific challenges. Unbalanced OT [25] relaxes the marginal constraints to compare distributions with unequal mass, which is particularly useful in biology and environmental modeling. Translation-invariant OT [26] incorporates invariance to spatial shifts, making it suitable for image processing and spatial pattern alignment. Diffusion OT [27] combines OT with diffusion processes, capturing geometric and structural properties in data, making it ideal for comparing data manifolds or temporally coherent patterns. These variations and computational methods have expanded OT's applicability to diverse domains, including weather prediction, single-cell gene expression analysis [19], and aviation decision support.

C. Related Work

Earthformer [28] shares a similar objective of leveraging Transformer and CNN architectures for modeling weather conditions. However, our approach diverges significantly in both model structure and the treatment of temporal dependencies. Earthformer employs a hierarchical encoder-decoder Transformer with cuboid attention, operating

in a non-autoregressive paradigm and using global tokens to coordinate information exchange between localized attention blocks. In contrast, our model utilizes a decoder-only Transformer design with fine-grained, frame-wise self-attention over spatial tokens. By incorporating causal masking, our architecture explicitly enforces temporal order, enabling autoregressive training where each output frame is based on past frames. This simplifies the architecture while preserving temporal consistency. Additionally, beyond image reconstruction, we incorporate Optimal Transport to compare latent representations, enabling retrievals of similar storm patterns. While OT has previously been applied to ensemble alignment [19], its integration with Vision Transformer-based embeddings remains largely unexplored. Our work bridges this gap by combining OT with learned latent spaces for semantically meaningful storm similarity analysis in aviation applications.

IV. Methodology

Our methodology combines Vision Transformers (ViTs) and Optimal Transport (OT) theory to analyze thunderstorm patterns for unlabelled data. This includes three stages: feature extraction, relationship analysis, and similarity measurement, providing a robust framework for clustering and understanding thunderstorms.

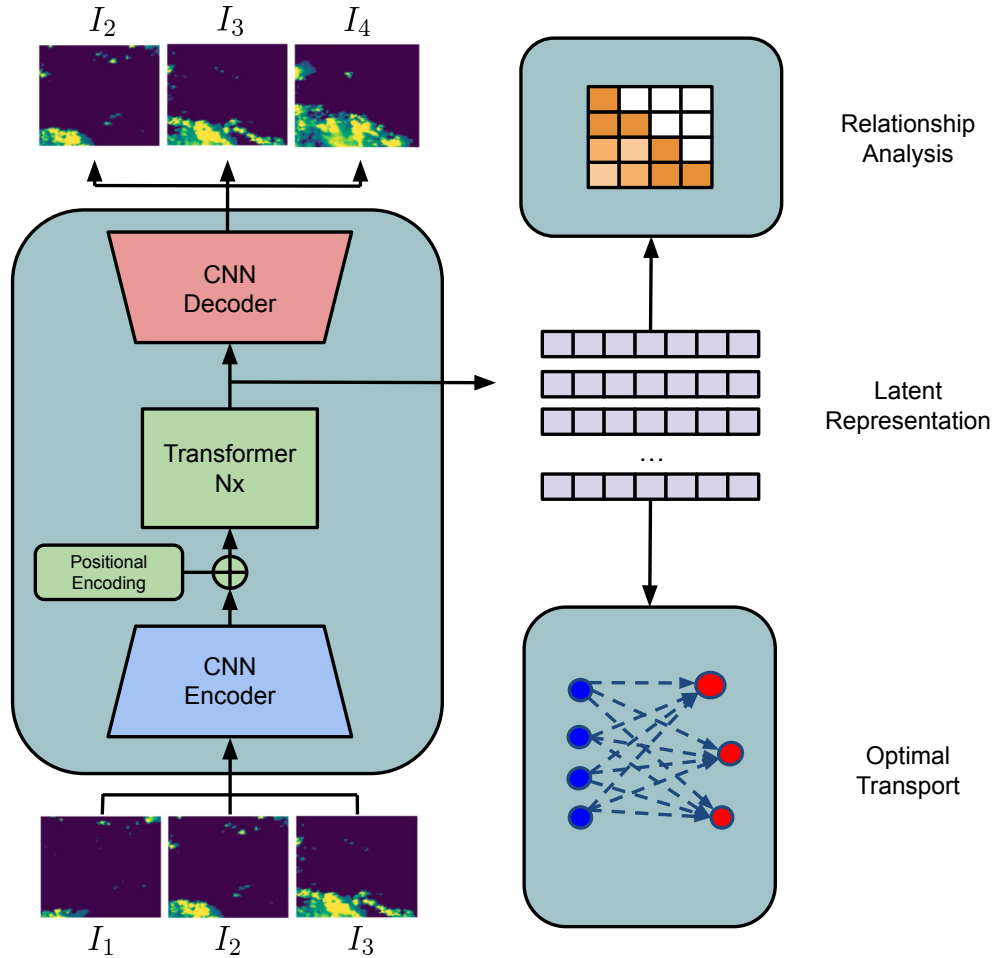


Fig. 1 Overview of our model architecture. The model combines convolutional neural networks (CNNs) with a transformer to effectively handle spatial-temporal weather data. A CNN encoder captures local spatial features from input images, which are then transformed into tokens with positional encodings. These tokens are processed through multiple transformer layers to model global spatial-temporal dependencies. A CNN decoder then reconstructs spatial outputs in an autoregressive style. The latent representation produced by the model is extracted for real-world applications such as relationship analysis or retrieval, guided by the OT theory.

A. Feature Extraction Using Vision Transformers

To construct a feature space that effectively encodes the dynamics of thunderstorms, we adopt an encoder-transformer-decoder architecture trained under an autoregressive paradigm. The model learns to predict future radar frames based on previous inputs, thereby encouraging the extraction of informative features that jointly represent both spatial structures and temporal transitions.

1. Vision Transformer Architecture

The input to our model is a sequence of radar images $[\mathbf{I}_1, \mathbf{I}_2, \dots, \mathbf{I}_T]$, where each frame $\mathbf{I}_t \in \mathbb{R}^{H \times W \times C}$ represents a radar image, measured by vertically integrated liquid (VIL) density or reflectivity, at time step t . Here, H and W denote the spatial dimensions of each frame, and C is the number of input channels. To extract spatial features, a shared **convolutional neural network (CNN) encoder** is independently applied to each frame in the sequence. The encoder compresses each frame into a compact feature tensor $\mathbf{Y}_t \in \mathbb{R}^{H' \times W' \times D}$, where H' and W' are the reduced spatial dimensions, and D is the number of feature channels. This encoder reduces spatial resolution while enriching semantic representation through deep convolutional processing.

To enable joint spatial-temporal modeling, each encoded feature tensor $\mathbf{Y}_t$ is reshaped into a sequence of tokens: $\mathbf{Y}_t \in \mathbb{R}^{N_p \times D}$, where $N_p = H' \times W'$ denotes the number of spatial tokens. These token sequences from every frames are concatenated across time to form the full input to the **transformer module**, denoted as $[\mathbf{Y}_1, \mathbf{Y}_2, \dots, \mathbf{Y}_T] \in \mathbb{R}^{N \times D}$, where $N = T \times N_p$ is the total number of tokens for the input sequence. The transformer processes this input using self-attention, capturing both intra-frame spatial relationships and inter-frame temporal dependencies, and outputs a latent representation $[\mathbf{Z}_1, \mathbf{Z}_2, \dots, \mathbf{Z}_T] \in \mathbb{R}^{N \times D}$ with the same dimensionality.

The latent representations $[\mathbf{Z}_1, \mathbf{Z}_2, \dots, \mathbf{Z}_T]$ are then fed into a shared **CNN decoder**, which reconstructs each frame from its corresponding token sequence. The decoder performs spatial upsampling to restore the original image resolution and channel structure, producing the output sequence $[\mathbf{O}_1, \mathbf{O}_2, \dots, \mathbf{O}_T]$ with each $\mathbf{O}_t \in \mathbb{R}^{H \times W \times C}$. This decoding stage is essential for autoregressive training, where the model generates future frames based on previous outputs while maintaining spatial coherence.

In summary, the architecture forms an end-to-end pipeline for spatial-temporal sequence modeling: the CNN encoder extracts per-frame spatial features, the transformer integrates temporal sequence, and the CNN decoder reconstructs the high-resolution output frames. The complete sequence transformation is represented as:

$$[\mathbf{I}_1, \dots, \mathbf{I}_T] \xrightarrow{\text{CNN encoder}} [\mathbf{Y}_1, \dots, \mathbf{Y}_T] \xrightarrow{\text{Transformer}} [\mathbf{Z}_1, \dots, \mathbf{Z}_T] \xrightarrow{\text{CNN decoder}} [\mathbf{O}_1, \dots, \mathbf{O}_T].$$

2. Autoregressive Training

The model is trained in an autoregressive manner using methods like masked autoencoding [21], which reconstruct missing patches and encourage the model to learn intrinsic spatial-temporal dependencies within the thunderstorm sequences. This embedding space effectively represents the spatial and temporal dynamics of storm systems. Specifically, the model is optimized by minimizing the mean squared error (MSE) between the model's predicted sequence $[\mathbf{O}_1, \mathbf{O}_2, \dots, \mathbf{O}_T]$ and the ground truth sequence that is shifted one time step forward $[\mathbf{I}_2, \mathbf{I}_3, \dots, \mathbf{I}_{T+1}]$ across all data sequences:

$$\mathcal{L}_{\text{recon}} = \frac{1}{M} \sum_{i=1}^M \sum_{t=1}^T \|\mathbf{O}_t - \mathbf{I}_{t+1}\|^2, \quad (5)$$

where M denotes the total number of sequential data. This objective guides the model to learn representations that are predictive of near-future storm dynamics.

To preserve causality, we apply a frame-wise masked attention mechanism, which restricts each token to attend *only* to tokens from the past or the same frame. This constraint prevents future information leakage and enforces consistency in sequence modeling.

B. Relationship Exploration Between Tokens and Images

The self-attention mechanism in the ViT provides a natural way to explore relationships between different tokens within images and across multiple images. The attention weights, α_{ij} , computed for each pair of patches i and j , reveal their importance:

$$\alpha_{ij} = \frac{\exp(\mathbf{q}_i \cdot \mathbf{k}_j / \sqrt{d_k})}{\sum_{j=1}^N \exp(\mathbf{q}_i \cdot \mathbf{k}_j / \sqrt{d_k})}. \quad (6)$$

By analyzing these weights, we identify patterns such as storm formation dynamics or localized regions of intensity. This relationship analysis is extended to compare images, exploring shared features between thunderstorms. Please see Section V for experimental results.

C. Similarity Measurement via Wasserstein Distance

The feature embeddings of each thunderstorm frame are treated as a probability distribution in the latent space. Let $\mathbf{A} = [\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_T]$ and $\mathbf{B} = [\mathbf{B}_1, \mathbf{B}_2, \dots, \mathbf{B}_T]$ denote the latent representations of two thunderstorm sequences. Each tensor $\mathbf{A}_t$ and $\mathbf{B}_t \in \mathbb{R}^{N_p \times D}$ corresponds to the latent feature representation of the frame at time t . From these tensors, token-wise distributions μ_t and ν_t are constructed to represent the feature embedding of each frame. To measure the similarity between these distributions, we compute their Wasserstein distance from the optimal transport theory:

$$W_p(\mu, \nu) = \left(\inf_{\gamma \in \Gamma(\mu, \nu)} \int \|x - y\|_p^p d\gamma(x, y) \right)^{1/p}, \quad (7)$$

where $\Gamma(\mu, \nu)$ denotes the set of all couplings (joint distributions) between $\mu(x)$ and $\nu(y)$. The Wasserstein distance quantifies the minimal cost of transporting one distribution into another, enabling a similarity measure that accounts for feature relationships. This frame-wise similarity is used to retrieve thunderstorm patterns that share analogous latent-space representations.

We extend this frame-wise similarity to the entire sequence by averaging the Wasserstein distances across time:

$$d_W(\mathbf{A}, \mathbf{B}) = \frac{1}{T} \sum_{t=1}^T W_p(\mu_t, \nu_t), \quad (8)$$

where W_p is computed between the empirical distributions μ_t and ν_t at each time step in the embedding space $\mathbb{R}^D$. This approach enables a temporally-informed comparison of thunderstorm sequences.

For implementation, we utilize the `ot.emd()` function from the Python Optimal Transport library to compute Wasserstein distances efficiently via linear programming. By incorporating the potential structure of token embeddings, Wasserstein distance offers a robust and sensitive metric for detecting spatial-temporal patterns in thunderstorm.

V. Experimental Results

A. Experimental Setup

1. Dataset Description

Our dataset consists of radar reflectivity images from the MULTI-RADAR/MULTI-SENSOR SYSTEM (MRMS) [29], covering a $300 \times 300 \text{ km}^2$ area centered on Dallas Fort Worth International Airport (DFW). Each radar frame has a spatial resolution of $3 \times 3 \text{ km}^2$ per pixel, resulting in images of size 100×100 . Data are collected at 10-minute intervals from March to October between 2016 and 2022, yielding a total of 205,848 frames. Reflectivity is used as the primary measure of thunderstorm intensity, with a commonly accepted threshold of 35 dBZ indicating storm activity [30].

2. Model Configuration and Parameters

The transformer includes 4 layers, each comprising multi-head self-attention (with 4 heads) and a feedforward block with hidden dimensionality of 32. A frame-wise causal masking strategy is used to preserve the order of intra-frame and inter-frame. The CNN encoder downsamples the input via two convolutional layers with stride-2, reducing a 100×100 input image to a 25×25 latent feature tensor. The decoder mirrors this structure with two transposed convolutions to restore spatial resolution.

The full list of training and model hyperparameters is summarized in Table 1.

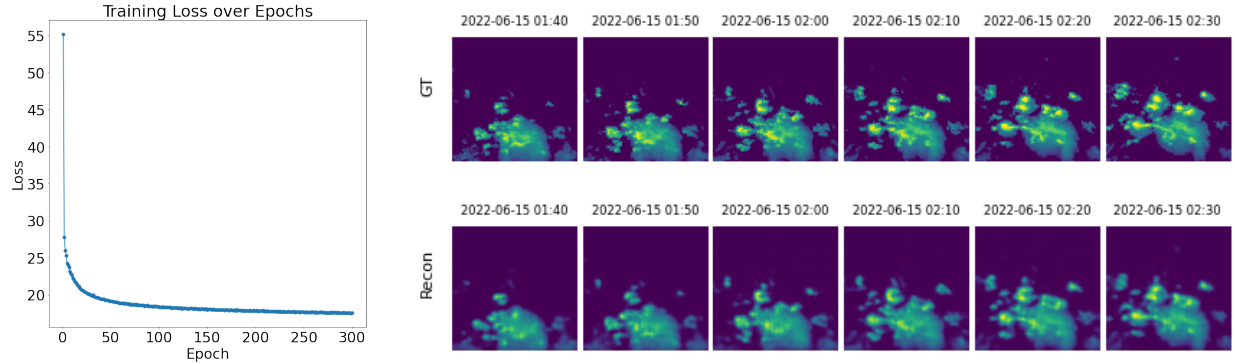
B. Reconstruction Performance

Following the setup described in Table 1, we train our models to fit the data. Figure 2(b) below demonstrates that our autoregressive model converges effectively after a sufficient number of training epochs. The output closely matches the ground truth from the test dataset, indicating high-resolution reconstruction quality. This suggests that the model

Table 1 Model Configuration and Training Parameters

Model Architecture	Value	Training Setup	Value
Feature Channels	8	Batch Size	8
Latent Resolution ($H' \times W'$)	25×25	Epochs	300
Image Sequence Length (Input/Output)	6 / 6	Learning Rate	0.001
Transformer Layers	4	Dropout Rate	0.1
Attention Heads	4	Test Ratio	0.2
Feedforward Hidden Dim	32		

successfully captures and reconstructs key features from the training data. As a result, the latent vector encapsulates key spatial patterns in a compact form, making it suitable not only for reconstruction but also for downstream tasks such as feature extraction.



(a) Training loss curve over 300 epochs, indicating convergence of our model.

(b) One example for ground truth vs. reconstructed radar reflectivity frames over time.

Fig. 2 Left: Convergence of training loss over epochs. Right: One example for reconstruction quality visualized through side-by-side comparison of predicted and actual radar observations.

C. Latent Space Visualization and Attention Relationship Exploration

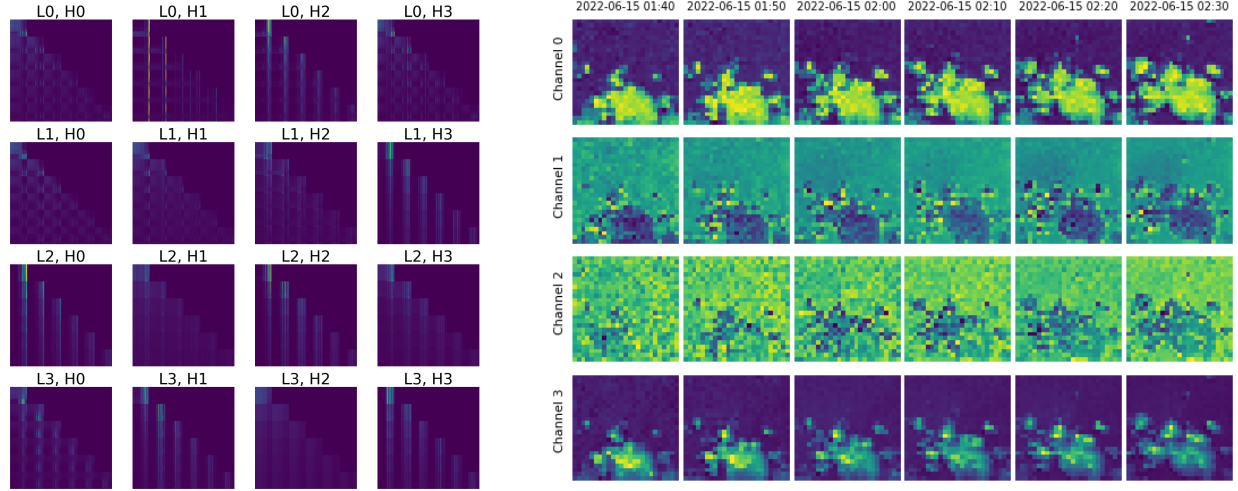
Using the reconstruction shown in Figure 2(b), we visualize the latent tensor produced after processing the input through the CNN encoder and transformer. As illustrated in Figure 3(b), the latent representation maintains the structural integrity of the original data at a coarser resolution, suggesting that the model effectively learns and encodes high-level features into a relatively lower space after sufficient training.

We further analyze the attention patterns among tokens over a sequence of frames, focusing on both “to” and “from” attention directions. Figure 4(a) illustrates how a particular token distributes its attention across other tokens throughout the sequence, whereas Figure 4(b) highlights which tokens direct their attention toward a specific token. In these visualizations, lighter pixels represent stronger attention weights (i.e., higher correlation), while darker pixels indicate weaker correlations.

These attention maps demonstrate the model’s ability to learn and retain complex spatial and temporal dependencies among tokens. This confirms that the model captures not only spatial relationships within each frame but also temporal dynamics across frames. As a result, the latent space provides a compact and expressive representation that effectively encodes both spatial and temporal information.

D. Similar Thunderstorm Retrieval

This section demonstrates a practical application of our model’s latent space for retrieving thunderstorms similar to a given reference event. Owing to the compact and semantically rich nature of the transformer-based latent representation,



(a) A full attention map illustrating the spatial-temporal relationships between tokens. The attention weights capture how different tokens in the thunderstorm sequence interact with the other tokens during the transformer process.

(b) Visualization of the first four channels from the latent tensor generated by the transformer. The preserved spatial patterns indicate that the CNN and transformer together effectively extract and encode high-level semantic features from the thunderstorm input.

Fig. 3 Latent space and attention map visualizations. These illustrate the model’s capability to learn and retain meaningful spatial-temporal representations of thunderstorm dynamics.

our method enables both efficient and accurate similarity searches.

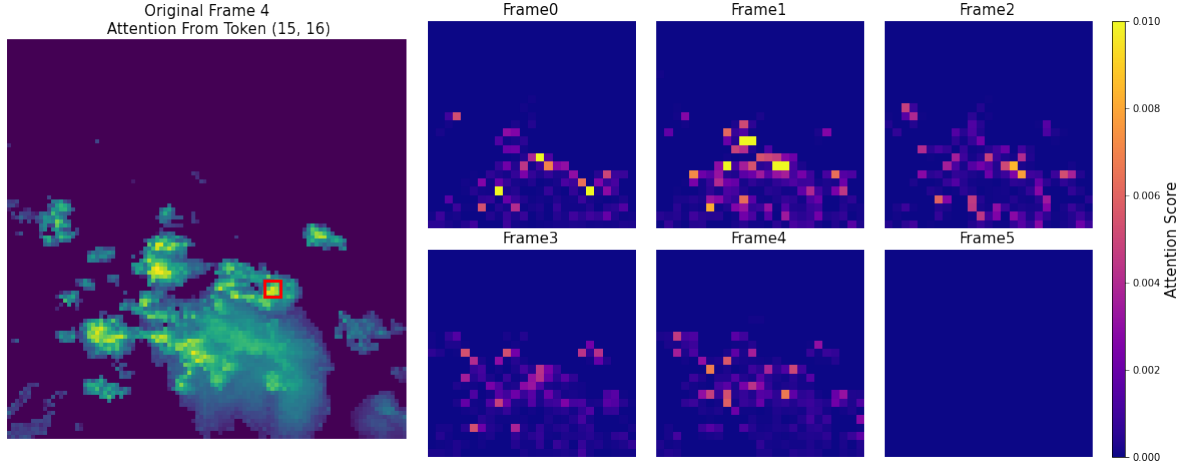
We evaluated the retrieval performance of the transformer’s latent space using three similarity metrics: cosine similarity, Euclidean distance, and token-wise Wasserstein distance, as detailed in Section IV. Both retrieval quality and computation time were analyzed to demonstrate the strengths of our approach. On a dataset of 115,540 sequences spanning 2018 to 2022, the average retrieval times were 19.82 seconds for cosine similarity, 9.29 seconds for Euclidean distance, and approximately 36 minutes for token-wise Wasserstein distance. Due to space constraints, we present only the top two most similar thunderstorm sequences retrieved by each method in Figure 5. These examples demonstrate that the latent representations effectively capture critical storm features, including motion trajectory, spatial structure, temporal evolution, shape, size, and intensity. Notably, all metrics show a strong preference for preserving temporal consistency. Cosine similarity and Euclidean distances generally emphasize spatial alignment, whereas the token-wise Wasserstein distance prioritizes shape similarity, even when spatial locations vary considerably. We believe that the most effective retrieval strategy is either combining multiple metrics or selecting the most appropriate one based on specific application needs.

VI. Disussion and Conclusion

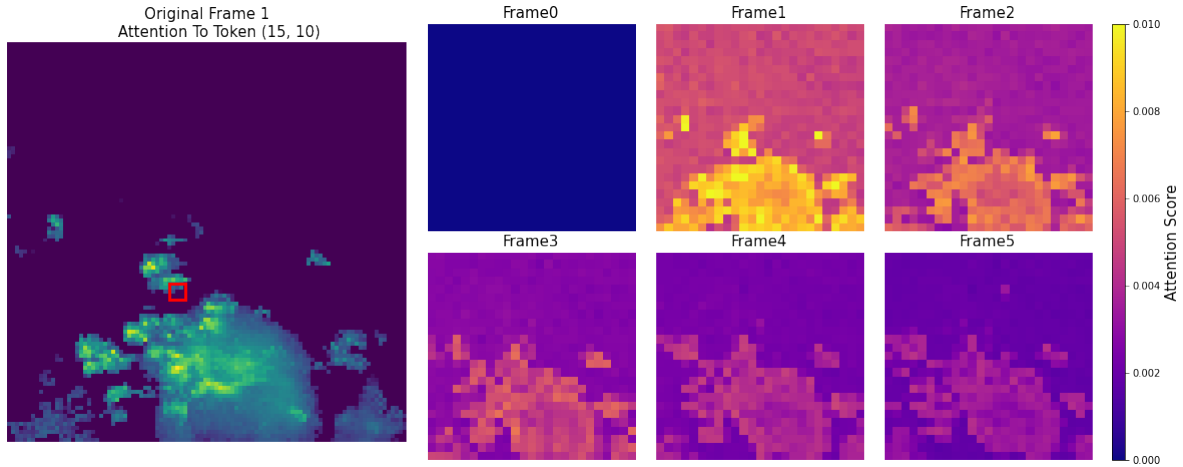
This study presents an integrated framework combining Vision Transformers (ViTs) and Optimal Transport (OT) theory to analyze thunderstorm sequences for aviation applications. By training the model autoregressively on radar reflectivity data, we extract latent embeddings that effectively capture both spatial structures and temporal dynamics of storm systems.

Attention visualizations confirm that the model learns complex relationships, including intra-frame and inter-frame, offering interpretability and insights into storm evolution. Compared to traditional similarity metrics, the Wasserstein distance enables more robust retrieval of structurally and temporally similar events. Experimental results demonstrate that our approach reconstructs storms with high fidelity and supports efficient similarity-based retrieval. These capabilities are valuable for pattern recognition, risk assessment, and decision-making in aviation meteorology.

Future work will expand the dataset to other regions, integrate additional meteorological features (e.g., wind speed, humidity), and explore probabilistic forecasting. We also aim to optimize OT computations and extend the framework to support real-time operations.



(a) From-attention map: indicates which tokens a specific token attends to across frames.



(b) To-attention map: indicates which tokens attend to a given token.

Fig. 4 Visualization of attention maps across a sequence of frames. Specific tokens are highlighted with a red box in the original frame on the left side. The attention maps for the latent tensor are shown on the right side.

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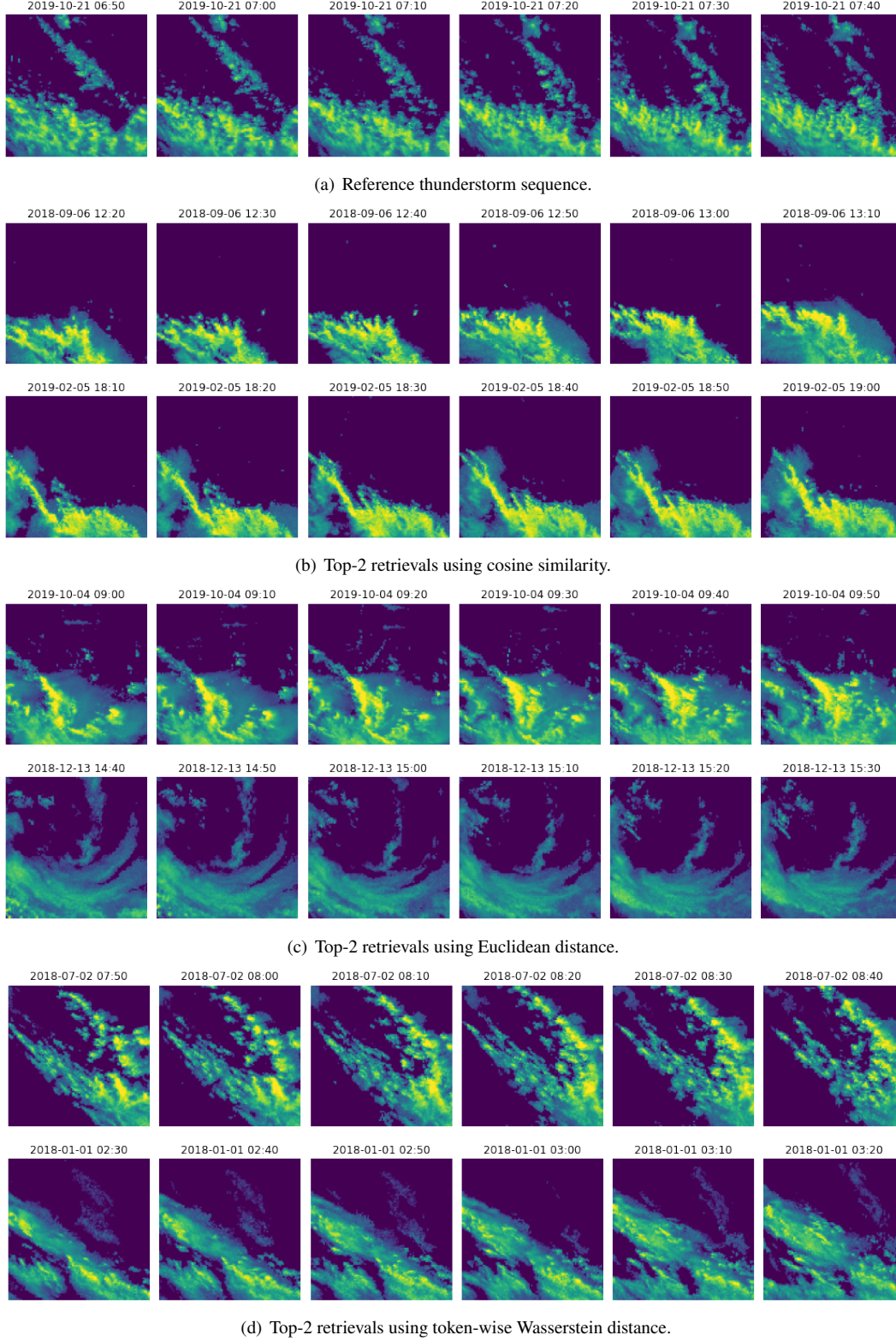


Fig. 5 Qualitative comparison of thunderstorm retrieval results using three similarity metrics in the transformer latent space. The top row presents the reference thunderstorm sequence, while each subsequent subfigure illustrates the top two sequences retrieved using the respective similarity metric. The retrieved examples demonstrate that the latent representation effectively captures key characteristics of the reference storms, including motion trajectory, spatial structure, temporal evolution, size, and intensity.